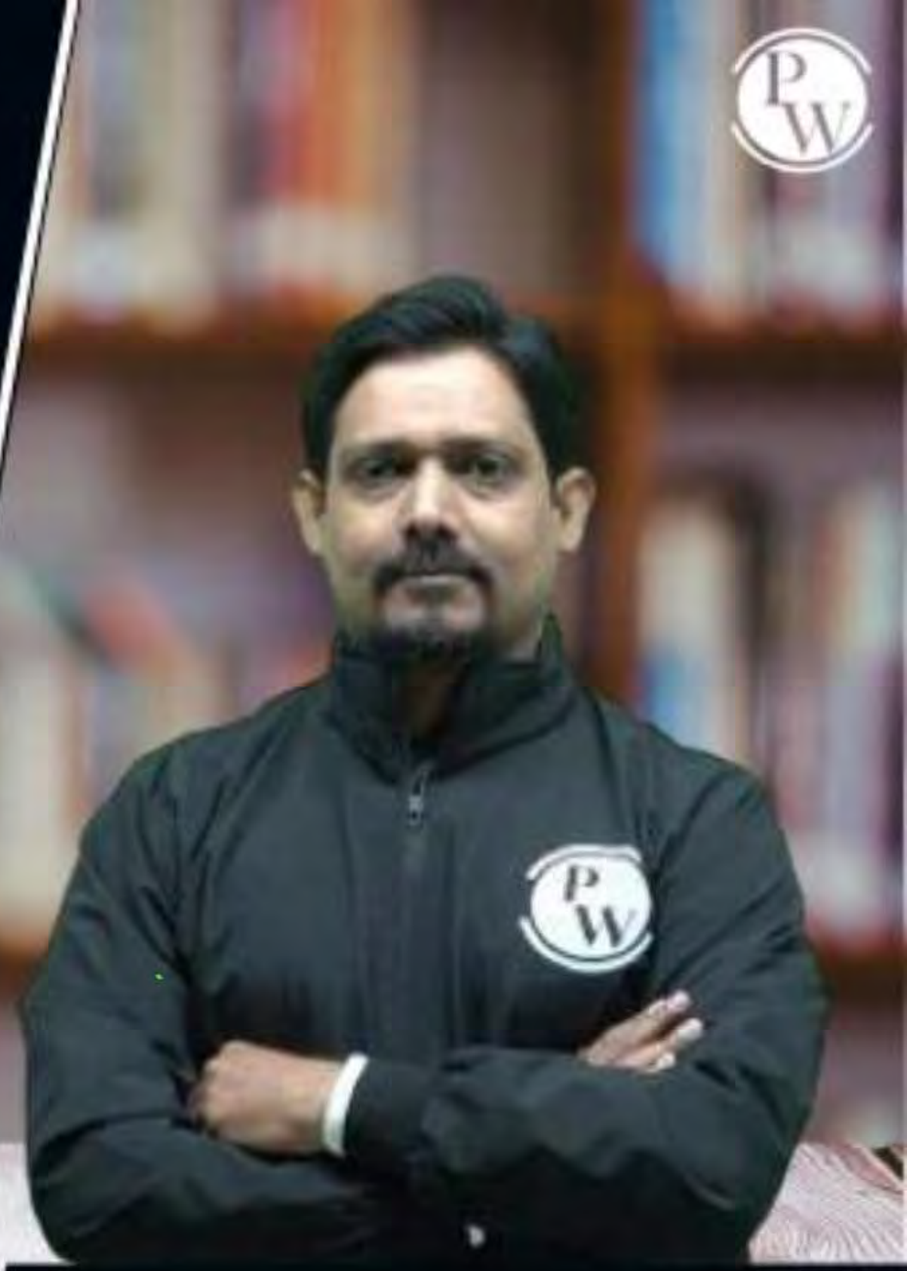
Computer Science & IT



Data Structure & Programming

Graph & Hashing

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic
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- Topic

Linear probing (primary clusters)

Quadratic probing

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Topics to be Covered



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Double Hashing

Slide



Topic : Hashing

Cycle Through the List

Key	$h(x,i) = (h(x), i^2) \bmod 10$
2	2
3	3
6	6
8	8
7	7
11	11%10 = 1
12	

12 will be mapped

Quadratic probing

$h(i) + i^2$

$h(i) = 12\%10 = 2$

$(2+1^2)\%10 = 3$

$(2+2^2)\%10 = 6$

$(2+3^2)\%10 = 1$

$(2+4^2)\%10 = 8$

$(2+5^2)\%10 = 7$

$(2+6^2)\%10 = 8$

$(2+7^2)\%10 = 1$

	0
11	1
2	2
3	3
	4
	5
6	6
7	7
8	8
	9

$(2+8^2) \bmod 10 = 6$ $(2+10^2)\%10 = 2$

$(2+9^2) \bmod 10$ 3

Slide



Topic : Hashing

* Same key value cycle Through the list
* Size of table m is prime No

Maximum Locahon Search



Slide



Topic : Hashing

Double Hashing

Here 2 Hash-function are present

primary Hash function → probe₀ $h_1(k)$

If the Location occupied → collision

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Topic : Hashing

The next probe location will be determine by Secondary Hash function

$h_2(k)$

$h_2(k)$ will determine offset value ← Not Location

Every key will have different offset value

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Topic : Hashing

18, 41, 22, 44, 59, 32, 31, 73

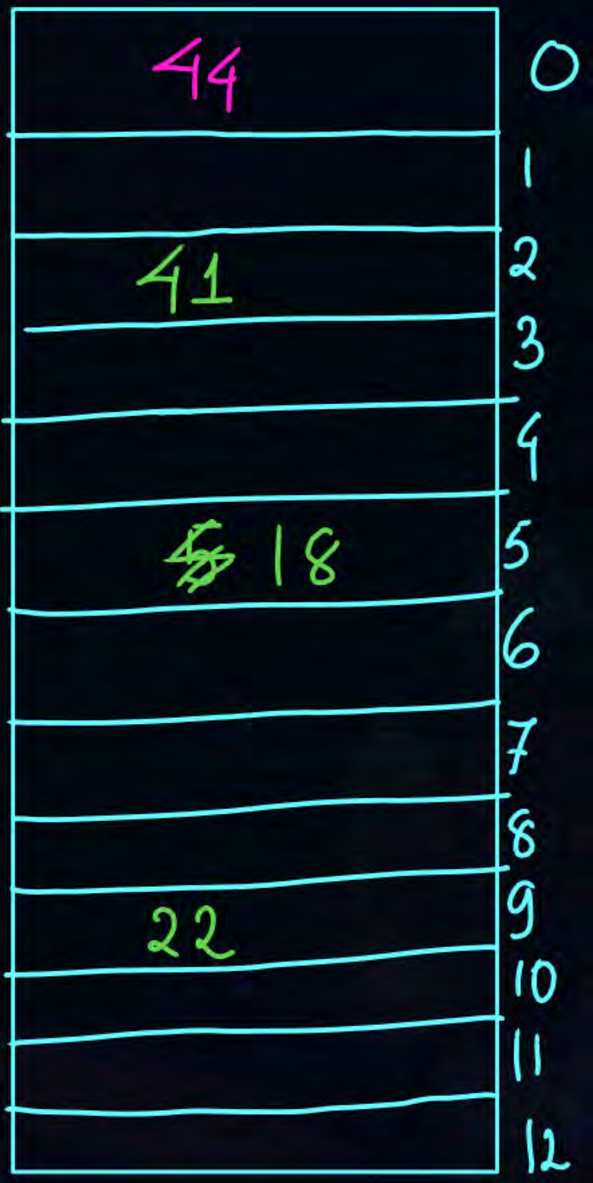
$h_1(x) = x \bmod 13$

$h_2(x) = 8 - (x \bmod 8)$ ← plan B

$h_1(44) = 44 \bmod 13 = 5$ ← probe 0

$h_2(44) = 8 - (44 \% 8)$

$= 8 - 4 = 4$



probe 1 =

$(5+4)\% 13$

$= 9$

probe 2

$= (9+4)\% 13$

$13 \% 13 = 0$

Slide

Topic : Hashing

18, 41, 22, 44, 59, 32, 31, 73

$h_1(x) = x \bmod 13$

$h_2(x) = 8 - (x \bmod 8)$ ← plan B

$h_1(59) \rightarrow 59\%13 = 7$

$h_1(32) \rightarrow 32\%13 = 6$

$h_1(31) \rightarrow 31\%13 = 5$



$h_2(31)$

$= 8 - (31\%8)$

$8 - 7 = 1$

$(5+1) = 6$ occupied

$(6+1) = 7$ occupied

$(7+1) = 8$ free

Slide



Topic : Hashing

18, 41, 22, 44, 59, 32, 31, 73

$h_1(x) = x \bmod 13$

$h_2(x) = 8 - (x \bmod 8)$ ← plan B

$h_1(73) = 73\%13 = 8$ probe 0

$h_2(73) = 8 - (73\%8)$

$= 8 - 1 = 7$

0	44
1	
2	41
3	73
4	
5	~~45~~ 18
6	32
7	59
8	31
9 ←	22
10	
11	
12	

probe 1

$= (8+7)\%13$

$= 15\%13 = 2$

$(2+7)\%13$

$= 9$ probe 2

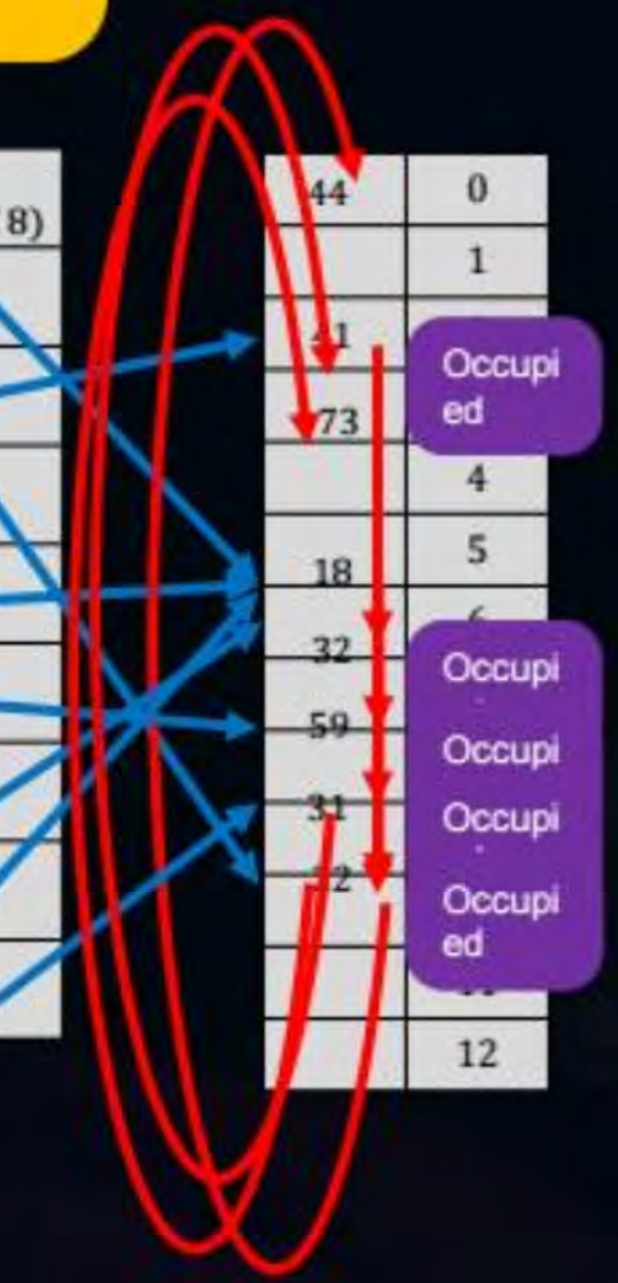
$(9+7)\%13$

$16\%13 = 3$

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Topic : Hashing

$Key - x$	$h_1(x) = x \bmod 13$	$h_2(x) = 8 - (x \bmod 8)$
18	5	6
41	2	7
22	9	2
44	5	4
59	7	5
32	6	8
31	5	1
73	8	7



Slide



Topic : Hashing

Consider a double hashing scheme in which the primary has function is $h_1(k) = k \bmod 23$, and the secondary hash function is $h_2(k) = 1 + (k \bmod 19)$. Assume that the table size is 23. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value k = 90 is

$h_1(k) = k \bmod 23$

$= 90 \bmod 23$

$= 21 \leftarrow$ probe 0

$h_2(90) = 1 + (90 \% 19)$

$= 1 + 14$

$= 15$ offset

90
76
14

probe 1 : $(21 + 15) \% 23 = 36 \% 23 = 13$

Slide



Topic : Hashing

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 10$, and the secondary hash function is $h_2(k) = 7 - (k \bmod 7)$. Assume that the table size is 10. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value k = 49 is

$h_1(k) = k \bmod 10$

$h_1 = 49 \bmod 10 = 9$

probe 0

$h_2(k) = 7 - (k \bmod 7)$

$h_2(49) = 7 - (49 \bmod 7)$

$= 7$

probe$_1$ = $(9+7)\%10$

$16\%10 = 6$

Slide



Topic : Hashing

Consider a double hashing scheme in which the primary hash function is $h_1(k) = k \bmod 10$, and the secondary hash function is $h_2(k) = 7 - (k \bmod 7)$. Assume that the table size is 10. Then the address returned by probe 1 in the probe sequence (assume that the probe sequence begins at probe 0) for key value $k = 49$ is **6**



2 mins Summary

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- Topic

Double Hashing

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THANK - YOU

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